

Cooperative Learning of Multi-Agent Systems Via Reinforcement Learning

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Abstract—In many specific scenarios, accurate and practical cooperative learning is a commonly encountered challenge in multi-agent systems. Thus, the current investigation focuses on cooperative learning algorithms for multi-agent systems and underpins an alternate data-based neural network reinforcement learning framework. To achieve the data-based learning optimization, the proposed cooperative learning framework, which comprises two layers, introduces a virtual learning objective. The followers learn the behaviors of the virtual objects in the first layer based on the adaptive neural networks (NNs). Specifically, the actor and critic NNs are applied to acquire cooperative behaviors and assess this layer's long-term utility function. Then another layer realizes the tracking performance between the virtual objects and the leader by introducing the local data-based performance index. Then, we formulate a resulting deterministic optimization problem and resolve it effectively with the policy iteration algorithm. This intuitive cooperative learning algorithm also preserves good robustness properties and eliminates the dependence on the prior knowledge of the multi-agent system model in the solution process. Finally, a multi-robot formation system demonstrates this promising development's practical appeal and highly effective outcome.

Index Terms—Neural networks, cooperative learning, multi-agent systems, reinforcement learning.

I. INTRODUCTION

COOPERATIVE learning of multi-agent systems facilitates an agent to perform objectives by interacting with its neighbor agents, which has encountered remarkable growth in past years and will continue to increase, given its capability of improving robustness and efficiency [1], [2], [3]. Cooperative learning plays a significant role in various fields, including intelligent transportation [4], aerospace systems [5], smart

grids [6], etc. Reinforcement learning, as one of the most practical learning branches, has attained substantial attention in the multi-agent systems' cooperative learning community due to its online learning framework and simplicity of implementation [7], [8], [9]. Besides, reinforcement learning concerns how agents shall select actions in an environment such that some concepts of accumulative reward are maximized, and the environment can be formulated as a Markov decision process. As a result, reinforcement learning shares better performance in cooperative behavior acquisition and in performing complex tasks, and some expected outcomes along this line are springing up [10], [11], [12], [13]. Reinforcement learning algorithms adopt sensible decisions rather than useless actions and achieve higher learning accuracy. However, how to tradeoff the exploration and exploitation in the cooperative behavior acquisition process decides the agents' learning efficiency. Deng et al. [12] proposed a novel learning framework that integrated reinforcement learning with knowledge transfer to improve learning performance. A Q-learning method was applied in [14] to avoid cooperation dilemmas and make all agents come to a joint agreement.

Note that the neural network has been widely used in imitating animal neural networks behavior characteristics and carrying out distributed parallel information processing. Artificial neural networks can be used but are not limited to data compression, image processing, and adaptive signal processing [15], [16], [17]. Furthermore, the universal approximation capability of the neural networks has particular merits in dealing with the cooperative learning of multi-agent systems with unknown nonlinear characteristics [18], [19], [20], which sheds light on the blossoming of cognitive, autonomous, and evolutionary robotics. However, one notes that almost all previous cooperative algorithms have been exclusively on the multi-agent systems in the desired communication channel despite the actuator faults and input dead zones being universal in practice. System faults will impact cooperative learning accuracy and even induce severe damage to a certain degree [21], [22]. Hence, the natural idea is to design NN-based fault-tolerant algorithms to eliminate the influence of actuator faults and dead zones and improve the multi-agent systems' robustness performance.

Recently, many researchers have formulated various representative algorithms to enrich the cooperative learning community of multi-agent systems [23], [24]. Unfortunately, the solution of the coupled Hamilton-Jacobi-Bellman (HJB) equation is generally impossible to obtain directly because the system dynamics are often difficult to model, and the external environment of

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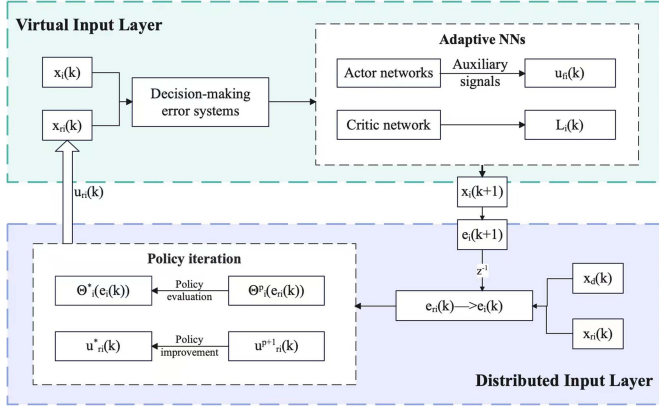


Fig. 1. Description of the cooperative learning framework.

the multi-agent systems is often uncertain [25]. In this case, a reasonable way is to assess optimization only with the system data instead of the accurate system model. Inspired by the above discussions, in the current investigation, we endeavor to develop a data-based NN cooperative learning algorithm for multi-agent systems via reinforcement learning. The contribution of this paper incorporates three aspects:

- We formulate a novel data-based optimal cooperative learning framework for a general class of discrete-time multi-agent systems with actuator faults and dead zones, which have been widely overlooked in previous contributions [26], [27], [28], [29].
- We established a two-layer learning framework via reinforcement learning. With this framework, system output and local input data are contained in the established performance function. Therefore, the data-based cooperative learning optimization can be achieved.
- We introduced two auxiliary signals to eliminate the impact of the actuator faults and dead zones. It shows better learning accuracy than the previous cooperative learning algorithms [30], which merged with the constraints of the bounded parameters and the known efficiency factors.

Fig. 1 describes the designed cooperative learning framework, which is composed of a virtual input layer and a distributed input layer. This framework integrates adaptive NNs and reinforcement learning methods to resolve the learning tasks. The organization of this paper is presented as follows: the graph theory concepts, the radial basis function neural networks (RBF NNs), and problem formulation are given in the following section, followed by the learning process and the performance analysis. Then, Section IV deals with the Bellman equation using the policy iteration (PI) algorithm. Finally, the simulation result is presented in Section V, and Section VI concludes this paper.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Graph Theory

A directed graph is introduced to express the information exchange among all the agents of the multi-agent system. The graph is represented by $\mathcal{G} = (\mathbb{V}, \mathbb{E}, \mathbb{A})$, $\mathbb{V} = (v_1, \dots, v_N)$ is a nonempty finite node set. $\mathbb{E} = \{e_{ij} = (v_i, v_j)\} \subset \mathbb{V} \times \mathbb{V}$

denotes the edge set. $\mathbb{A} = [a_{ij}]_{N \times N}$ represents the weighted adjacency matrix with $a_{ij} > 0$ indicating agent j can communicate with agent i , otherwise, $a_{ij} = 0$. Without loss of generality, assume $a_{ii} = 0$. $\mathbb{N}_i = \{j : (v_j, v_i) \in \mathbb{E}\}$ represents the neighbor set of agent i . $\mathbb{D} = \text{diag}\{h_1, \dots, h_N\}$ implies the in-degree matrix, where $h_i = \sum_{j \in \mathbb{N}_i} a_{ij}$. The digraph Laplacian matrix $\mathbb{L}$ is derived as $\mathbb{L} = \mathbb{D} - \mathbb{A}$. Define the matrix $H = \text{diag}\{h_1, \dots, h_N\}$ representing the connection matrix between followers and the leader, where $h_i > 0$ implies the i th follower can communicate with the leader, otherwise $h_i = 0$.

B. Problem Formulation

Consider a general class of multi-agent systems containing N followers and one leader, where each follower's dynamics is modeled as:

$$\begin{cases} x_{i,l}(k+1) = b_{i,l}x_{i,l+1}(k) + f_{i,l}(\bar{x}_{i,l}(k)), \\ x_{i,n}(k+1) = b_{i,n}u_{fi}(k) + f_{i,n}(x_i(k)) + d_i(k) \end{cases} \quad (1)$$

where $i = 1, 2, \dots, N$, $l = 1, 2, \dots, n-1$, $\bar{x}_{i,l}(k) = [x_{i,1}(k), \dots, x_{i,l}(k)]^T \in \mathbb{R}^l$ stands for the state vector. Define $x_i(k) = \bar{x}_{i,n}(k)$. $f_{i,j}(\cdot)$ represents the unknown smooth nonlinear characteristic of the follower's state. $b_{i,j}$ refers to as a parameter to be specified subsequently, and $d_i(k)$ is an unknown external disturbance satisfying $|d_i(k)| < \bar{d}$ with $\bar{d}$ being a positive constant. $u_{fi}(k) \in \mathbb{R}$ indicates the system input. The output of each agent is $x_{i,1}(k)$. For convenience, the representation $f_{i,j}(\bar{x}_{i,j}(k))$ is replaced by $f_{i,j}(k)$. Let the bounded signal $x_0(k) \in \mathbb{R}$ be the state output of the leader.

In practical engineering systems, the control signals may face attenuation and floating. Motivated by the propositions in [31], [32], [33], the actuator fault in system (1) can be modeled as $u_{fi}(k) = \eta_i(k)u_i(k) + \delta_i(k)$, where $\eta_i(k)$ and $\delta_i(k)$ denote the efficiency factor and the unknown drift of the actuator, respectively. The control signal may be discounted during the transmission procedure. Thus, regarding the efficiency factor, $0 < \underline{\eta} < |\eta_i(k)| < \bar{\eta} \leq 1$ is held. Besides, suppose that the bias fault is bounded, one has $|\delta_i(k)| \leq \bar{\delta}$ with $\bar{\delta}$ being a positive scalar.

In practical industry, dead zone is widely used to avoid oscillation. To show the generality of the multi-agent systems considered, suppose the dead zone exists, which means the control signals may face value changes or disappearances within the specific scope. Then, $u_i(k)$ can be formulated as $u_i(k) = D(v_i(k))$, which is defined as

$$D(v_i(k)) = \begin{cases} g_{i,r}(v_i(k) - t_{i,r}), & v_i(k) \geq t_{i,r} \\ 0, & -t_{i,l} < v_i(k) < t_{i,r} \\ g_{i,l}(v_i(k) + t_{i,l}), & v_i(k) \leq -t_{i,l} \end{cases} \quad (2)$$

where $v_i(k)$ is the real input signal without any faults. The right and left slopes of the dead zone are $g_{i,r}$ and $g_{i,l}$, respectively. $t_{i,r}$ and $t_{i,l}$ stand for the nonlinearity breakpoints. $g_{i,r}$, $g_{i,l}$, $t_{i,r}$ and $t_{i,l}$ are positive constants for $i = 1, \dots, N$. According to the pioneering contributions in [2], $D(v_i(k))$ can be expressed as $D(v_i(k)) =$

$$g_i(k)v_i(k) + t_i(k), \text{ where } g_i(k) = \begin{cases} g_{i,r}, & v_i(k) > 0 \\ g_{i,l}, & v_i(k) \leq 0 \end{cases}, \text{ and}$$

$$t_i(k) = \begin{cases} -g_{i,r}t_{i,r}, & v_i(k) \geq t_{i,r} \\ -g_i(k)v_i(k), & -t_{i,l} < v_i(k) < t_{i,r} \\ g_{i,l}t_{i,l}, & v_i(k) \leq -t_{i,l} \end{cases}. \text{ Accordingly,}$$

one has $\underline{g} = \min\{g_{i,r}, g_{i,l}\} \leq |g_i(k)| \leq \bar{g} = \max\{g_{i,r}, g_{i,l}\}$ and $|t_i(k)| \leq \bar{t} = \max\{g_{i,r}t_{i,r}, g_{i,l}t_{i,l}\}$ for $\forall i$.

In virtue of the multi-agent system theory, define the local tracking error $e_i(k) = \sum_{j \in \mathbb{N}_i} a_{ij}(x_{i,1}(k) - x_{j,1}(k)) + h_i(x_{i,1}(k) - x_0(k))$. Based on the results in [34], as long as the communication topology has a spanning tree, one has if $\|e_i(k)\| \rightarrow 0$, then $\|x_{i,1}(k) - x_0(k)\| \rightarrow 0$, i.e., all the agents will reach synchronization with the leader when local tracking errors converge to 0.

This paper explores the problems of energy consumption optimization and cooperative learning simultaneously. Concerning the dynamic analysis of a system, it is usually favored that certain performance restraints should be satisfied. For example, we shall minimize the cost function under the preferred input intent. In light of the optimal principle, this paper formulates the local performance index function of each follower i as

$$J_i(e_i(k), \mu_i(k), \mu_{\mathbb{N}_i}(k)) = \sum_{l=k}^{\infty} \gamma^{l-k} r_i(e_i(l), \mu_i(l), \mu_{\mathbb{N}_i}(l)) \quad (3)$$

where $r_i(e_i(l), \mu_i(l), \mu_{\mathbb{N}_i}(l)) = e_i(l)^T Q_{ii} e_i(l) + \mu_i^T(l) R_{ii} \mu_i(l) + \mu_{\mathbb{N}_i}^T(l) R_{ij} \mu_{\mathbb{N}_i}(l)$ with Q_{ii} , R_{ii} and R_{ij} being positive definite matrices. $\mu_i(k)$ and $\mu_{\mathbb{N}_i}(k)$ denote the practical system input of the i th follower and its underlying neighbor agents. γ represents the discount factor satisfying $0 < \gamma \leq 1$.

C. Radial Basis Function Neural Networks (RBF NNs)

The RBF NN [35], [36] is formulated as $F_{nn}(Z) = W^T \Phi(Z)$, where $W = [W_1, \dots, W_\iota]^T \in \mathbb{R}^\iota$ denotes the ideal weight vector satisfying $\|W\| \leq \bar{W}$ with $\bar{W}$ being a positive constant. ι represents the number of the neurons in the hidden layer. $\Phi(Z) = [\Phi_1(Z), \dots, \Phi_\iota(Z)]^T \in \mathbb{R}^\iota$ is the bias vector. The basis function can be selected as $\Phi_m(Z) = \exp(-\frac{(Z-\zeta_m)^T(Z-\zeta_m)}{2\rho_m^2})$, in which ζ_m is the center of the receptive field and ρ_m is the width of Gaussian kernel for $m = 1, \dots, \iota$. Obviously, one has $0 < \Phi^T(Z)\Phi(Z) \leq \iota$. That is to say, there exists an RBF such that $F(Z) = W^T \Phi(Z) + \varepsilon(Z)$, where W and $\varepsilon(Z)$ stand for the ideal weights and the approximation error, respectively. It is supposed that $|\varepsilon(Z)| \leq \bar{\varepsilon}$ with $\bar{\varepsilon}$ being positive and unknown.

The used assumption and definition in this paper are listed as follow.

Assumption 1: It is supposed that the multi-agent system communication digraph we study contains a spanning tree, which means there exists a directed path from the leader to any other follower agent.

Definition 1 (Admissible Control [37]): An input policy $u_i(k)$ is defined to be admissible on a compact set $\Omega_x \in \mathbb{R}^n$ if $u_i(0) =$

0, $u_i(k)$ is continuous, and $u_i(k)$ can stabilize the system it controls.

III. VIRTUAL LEARNING DESIGN WITH ADAPTIVE NNs

In this section, a virtual learning objective is introduced and the cooperative learning framework which consists of two layers is designed. The followers learn the behavior of the virtual objects in the first layer based on the adaptive neural networks. For proceeding, the virtual learning objective for the i th follower is designed as

$$\begin{cases} x_{ri1}(k+1) = A_{i1}\bar{x}_{ri1}(k) + b_{i,1}x_{ri2}(k) \\ x_{ril}(k+1) = A_{il}\bar{x}_{ril}(k) + b_{i,l}x_{ri(l+1)}(k) \\ x_{rin}(k+1) = A_{in}\bar{x}_{rin}(k) + b_{i,n}u_{ri}(k) \end{cases} \quad (4)$$

where $l = 2, \dots, n-1$, $\bar{x}_{ril}(k) = [x_{ri1}(k), \dots, x_{ril}(k)]^T \in \mathbb{R}^l$ denotes the virtual learning objective. Define $x_{ri}(k) = \bar{x}_{rin}(k)$. $u_{ri}(k) \in \mathbb{R}$ is the virtual input specified later. $A_{il} \in \mathbb{R}^{1 \times l}$, $l = 1, \dots, n$ is a system matrix. Define the tracking error between the system (1) and the virtual learning objective (4) as $\tilde{x}_{ri}(k)$, which is formulated by $\tilde{x}_{ri}(k) = x_i(k) - x_{ri}(k)$.

A. Critic Neural Networks

Define $e_{i,1}(k) = \tilde{x}_{ri1}(k)$. Then the utility function of the tracking error system $\tilde{x}_{ri}(k)$ is defined as

$$Q_i(k) = \begin{cases} 0, & |e_{i,1}(k)| < c_1 \\ 1, & |e_{i,1}(k)| \geq c_1 \end{cases},$$

where c_1 and $Q_i(k)$ refer to as the threshold and the index of the tracking performance, respectively.

Let $L_i(k)$ be the long-term utility function to measure the learning performance. It can be formulated as $L_i(k) = Q_i(k+1) + \pi_i Q_i(k+2) + \pi_i^2 Q_i(k+3) + \dots$, where π_i stands for the discount factor with $0 < \pi_i < 1$. With the learning capability of the critic NNs, it yields $L_i(k) = W_{i,c}^T \Phi_{i,c}(k) + \varepsilon_{i,c}(k)$ where $W_{i,c}$ is the ideal weight vector. $\varepsilon_{i,c}(k)$ represents the approximation error. $\Phi_{i,c}(k)$ denotes the bias vector. Suppose $\|W_{i,c}\| \leq \bar{W}_c$ and $|\varepsilon_{i,c}(k)| \leq \bar{\varepsilon}_c$ with $\bar{W}_c$ and $\bar{\varepsilon}_c$ being positive constants.

However, the ideal weight vector is difficult, even impossible to measure. Thus, the estimate of $L_i(k)$ can be described as $\hat{L}_i(k) = \hat{W}_{i,c}^T(k) \Phi_{i,c}(k)$ where $\hat{W}_{i,c}(k)$ is the estimate of $W_{i,c}$. The Bellman error can be defined as $e_{i,c}(k) = \pi_i \hat{L}_i(k) - [\hat{L}_i(k-1) - Q_i(k)]$.

During the critic network training procedure, to minimize the objective cost function $E_{i,c}(k) = \frac{1}{2} e_{i,c}^2(k)$, the standard gradient-descent-based method is utilized to adjust the weights. Define the approximation error as $\tilde{W}_{i,c}(k) = \hat{W}_{i,c}(k) - W_{i,c}$. The updating law of $\hat{W}_{i,c}(k)$ is

$$\hat{W}_{i,c}(k+1) = \hat{W}_{i,c}(k) - \lambda_{i,c} \pi_i \Phi_{i,c}(k) e_{i,c}(k) \quad (5)$$

with $\lambda_{i,c} > 0$ being the learning rate.

B. Actor Neural Networks

In what follows, combining with the backstepping manner, the adaptive weight learning laws based on actor NNs are developed. The detailed design procedure is given as follows.

Step 1: Define $e_{i,2}(k+n-1) = \tilde{x}_{ri2}(k+n-1) - \alpha_{i,1}(k)$ where $\alpha_{i,1}(k)$ is the virtual learning controller for $\tilde{x}_{ri1}(k)$. Then, $e_{i,1}(k+n)$ can be derived as

$$e_{i,1}(k+n) = \mathcal{F}_{i,1}(\bar{x}_{i,n}(k)) - A_{i1}\bar{x}_{ri1}(k+n-1) + b_{i,1}(e_{i,2}(k+n-1) + \alpha_{i,1}(k)) \quad (6)$$

where $\mathcal{F}_{i,1}(\bar{x}_{i,n}(k))$ is an unknown function.

Design $\alpha_{i,1}(k)$ as $\alpha_{i,1}(k) = -\frac{1}{b_{i,1}}(\mathcal{F}_{i,1}(\bar{x}_{i,n}(k)) - A_{i1}\bar{x}_{ri1}(k+n-1))$. Since $\mathcal{F}_{i,1}(\cdot)$ is unknown, $\alpha_{i,1}(k)$ can be approximated by the actor NN as $\alpha_{i,1}(k) = W_{i,1}^T \Phi_{i,1}(k) + \varepsilon_{i,1}(k)$ where $\|W_{i,1}\| \leq \bar{W}_{i,1}$ and $|\varepsilon_{i,1}(k)| \leq \bar{\varepsilon}_{i,1}$, with $\bar{W}_{i,1}$ and $\bar{\varepsilon}_{i,1}$ being positive constants. Practical $\hat{\alpha}_{i,1}(k)$ is derived as $\hat{\alpha}_{i,1}(k) = \hat{W}_{i,1}(k)^T \Phi_{i,1}(k)$ where $\hat{W}_{i,1}(k)$ is an estimate of $W_{i,1}$.

Then, one has $e_{i,1}(k+n) = b_{i,1}(e_{i,2}(k+n-1) + \hat{\alpha}_{i,1}(k) - \alpha_{i,1}(k))$. Substituting $\hat{\alpha}_{i,1}(k)$ into $e_{i,1}(k+n)$, yields

$$e_{i,1}(k+1) = b_{i,1}(e_{i,2}(k) + \tilde{W}_{i,1}^T(k_1)\Phi_{i,1}(k_1) - \varepsilon_{i,1}(k_1)) \quad (7)$$

where $k_1 = k - n + 1$ and $\tilde{W}_{i,1}(k) = \hat{W}_{i,1}(k) - W_{i,1}$.

Define the strategic utility function as $O_{i,1}(k) = \tilde{W}_{i,1}^T(k_1)\Phi_{i,1}(k_1) + [\hat{L}_i(k) - L_d(k)]$, where $L_d(k) = 0$ represents the expected utility function.

Choose the cost function as $E_{i,1}(k) = \frac{1}{2}O_{i,1}^2(k)$. The updated law of $\hat{W}_{i,1}(k)$ is derived with the gradient descent method:

$$\begin{aligned} \hat{W}_{i,1}(k+1) &= \hat{W}_{i,1}(k_1) - \lambda_{i,1}\Phi_{i,1}(k_1) \\ &\quad \times [\hat{W}_{i,1}^T(k_1)\Phi_{i,1}(k_1) + \hat{L}_i(k)] \end{aligned} \quad (8)$$

where $\lambda_{i,1} > 0$ represents the learning rate.

Step l ($2 \leq l \leq n-1$): The learning error in step l can be described as $e_{i,l}(k+n-l+1) = \tilde{x}_{ril}(k+n-l+1) - \alpha_{i,l-1}(k)$. Then, one obtains

$$\begin{aligned} e_{i,l}(k+n-l+1) &= \mathcal{F}_{i,l}(\bar{x}_{i,n}(k)) - A_{il}\bar{x}_{ril}(k+n-l) \\ &\quad + b_{i,l}\tilde{x}_{ri(l+1)}(k+n-l) - \alpha_{i,l-1}(k). \end{aligned} \quad (9)$$

where $\mathcal{F}_{i,l}(\bar{x}_{i,n}(k))$ is an unknown function.

Design the virtual input $\alpha_{i,l}(k)$ for $\tilde{x}_{ril}(k)$ as

$$\begin{aligned} \alpha_{i,l}(k) &= -\frac{1}{b_{i,l}}(\mathcal{F}_{i,l}(\bar{x}_{i,n}(k)) \\ &\quad - A_{il}\bar{x}_{ril}(k+n-l) + \alpha_{i,l-1}(k)). \end{aligned} \quad (10)$$

Accordingly, one has $\alpha_{i,l}(k) = W_{i,l}^T \Phi_{i,l}(k) + \varepsilon_{i,l}(k)$, where $\|W_{i,l}\| \leq \bar{W}_{i,l}$ and $|\varepsilon_{i,l}(k)| \leq \bar{\varepsilon}_{i,l}$ with $\bar{W}_{i,l}$ and $\bar{\varepsilon}_{i,l}$ being positive constants. The approximation of $\alpha_{i,l}(k)$ is defined as $\hat{\alpha}_{i,l}(k) = \hat{W}_{i,l}(k)^T \Phi_{i,l}(k)$, where $\hat{W}_{i,l}(k)$ refers to as the estimation of $W_{i,l}$.

Let the tracking error of the next step be $e_{i,l+1}(k+n-l) = \tilde{x}_{ri(l+1)}(k+n-l) - \alpha_{i,l}(k)$ and apply the similar manner in step 1, one has

$$e_{i,l}(k+1) = b_{i,l}(e_{i,l+1}(k) + \tilde{W}_{i,l}^T(k_l)\Phi_{i,l}(k_l) - \varepsilon_{i,l}(k_l)) \quad (11)$$

where $k_l = k - n + l$ and $\tilde{W}_{i,l}(k) = \hat{W}_{i,l}(k) - W_{i,l}$.

Define the strategic utility function as $O_{i,l}(k) = \tilde{W}_{i,l}^T(k_l)\Phi_{i,l}(k_l) + [\hat{L}_i(k) - L_d(k)]$. Choose the cost function as $E_{i,l}(k) = \frac{1}{2}O_{i,l}^2(k)$. In the gradient descent manner, the updating law of $\hat{W}_{i,l}(k)$ is designed as:

$$\begin{aligned} \hat{W}_{i,l}(k+1) &= \hat{W}_{i,l}(k_l) - \lambda_{i,l}\Phi_{i,l}(k_l) \\ &\quad \times [\hat{W}_{i,l}^T(k_l)\Phi_{i,l}(k_l) + \hat{L}_i(k)] \end{aligned} \quad (12)$$

where $\lambda_{i,l} > 0$ indicates the learning rate.

Step n : Define $e_{i,n}(k+1) = \tilde{x}_{rin}(k+1) - \alpha_{i,n-1}(k)$, where $\alpha_{i,n-1}(k)$ represents the virtual signal in the $n-1$ step. From system $\tilde{x}_{ri}(k)$, one can derive

$$\begin{aligned} e_{i,n}(k+1) &= \mathcal{F}_{i,n}(\bar{x}_{i,n}(k)) - A_{in}\bar{x}_{rin}(k) + d_i(k) \\ &\quad + b_{i,n}(\eta_i(k)(g_i(k)v_i(k) + t_i(k)) + \delta_i(k)) \\ &\quad - b_{i,n}u_{ri}(k) - \alpha_{i,n-1}(k). \end{aligned} \quad (13)$$

where $\mathcal{F}_{i,n}(\bar{x}_{i,n}(k))$ is an unknown function.

Design the virtual input $\alpha_{i,n}(k)$ for $\tilde{x}_{rin}(k)$ as

$$\begin{aligned} \alpha_{i,n}(k) &= -\frac{1}{b_{i,n}g_i(k)\eta_i(k)}(\mathcal{F}_{i,n}(\bar{x}_{i,n}(k)) \\ &\quad - A_{in}\bar{x}_{rin}(k) - b_{i,n}u_{ri}(k) - \alpha_{i,n-1}(k)). \end{aligned} \quad (14)$$

It can be approximated by the RBF NN that $\alpha_{i,n}(k) = W_{i,n}^T \Phi_{i,n}(k) + \varepsilon_{i,n}(k)$ where $\|W_{i,n}\| \leq \bar{W}_{i,n}$ and $|\varepsilon_{i,n}(k)| \leq \bar{\varepsilon}_{i,n}$ with $\bar{W}_{i,n}$ and $\bar{\varepsilon}_{i,n}$ being positive constants. It follows that

$$\begin{aligned} e_{i,n}(k+1) &= b_{i,n}\eta_i(k)g_i(k)(v_i(k) + \frac{t_i(k)}{g_i(k)} \\ &\quad + \frac{\delta_i(k)}{g_i(k)\eta_i(k)} - \alpha_{i,n}(k)) + d_i(k). \end{aligned} \quad (15)$$

Choose two unknown variables ξ_i and Γ_i to represent the upper bounds of $\frac{t_i(k)}{g_i(k)}$ and $\frac{\delta_i(k)}{g_i(k)\eta_i(k)}$. Owing to the fact $t_i(k)$, $g_i(k)$, $\eta_i(k)$ and $\delta_i(k)$ are bounded, one has $\xi_i \triangleq \frac{\bar{t}}{\underline{g}}$ and $\Gamma_i \triangleq \frac{\bar{\delta}}{\underline{g}\eta}$.

Choose $v_i(k) = \alpha_{i,n}(k) + \xi_i + \Gamma_i$, the estimate of $v_i(k)$ is

$$\hat{v}_i(k) = \hat{W}_{i,n}^T(k)\Phi_{i,n}(k) + \hat{\xi}_i(k) + \hat{\Gamma}_i(k). \quad (16)$$

Then, one has

$$\begin{aligned} e_{i,n}(k+1) &= b_{i,n}\eta_i(k)g_i(k)(\tilde{W}_{i,n}^T(k_n)\Phi_{i,n}(k_n) + \hat{\xi}_i(k) \\ &\quad + \hat{\Gamma}_i(k) \\ &\quad - \varepsilon_{i,n}(k_n)) + b_{i,n}\eta_i(k)g_i(k)\left(\frac{t_i(k)}{g_i(k)} + \frac{\delta_i(k)}{g_i(k)\eta_i(k)}\right) + d_i(k) \end{aligned}$$

where $k_n = k$ and $\tilde{W}_{i,n}(k) = \hat{W}_{i,n}(k) - W_{i,n}$.

The strategic utility function is defined as $O_{i,n}(k) = \tilde{W}_{i,n}^T(k_n)\Phi_{i,n}(k_n) + [\hat{L}_i(k) - L_d(k)]$. Combining with the

cost function $E_{i,n}(k) = \frac{1}{2}O_{i,n}^2(k)$ and the gradient descent method, the updating law of $\hat{W}_{i,n}(k)$ can be derived as:

$$\begin{aligned} \hat{W}_{i,n}(k+1) &= \hat{W}_{i,n}(k_n) - \lambda_{i,n}\Phi_{i,n}(k_n) \\ &\times \left[\hat{W}_{i,n}^T(k_n)\Phi_{i,n}(k_n) + \hat{L}_i(k) \right] \end{aligned} \quad (17)$$

where $\lambda_{i,n} > 0$ is the learning rate.

Note that the unknown auxiliary signals $\hat{\xi}_i(k)$ and $\hat{\Gamma}_i(k)$ shall be updated to make all the followers reach consensus with the virtual objective, hence, let the utility function be

$$\begin{aligned} O_{i,\xi}(k) &= \hat{\xi}_i(k) + [\hat{L}_i(k) - L_d(k)] \\ O_{i,\Gamma}(k) &= \hat{\Gamma}_i(k) + [\hat{L}_i(k) - L_d(k)]. \end{aligned} \quad (18)$$

Similar as the other cost functions, the auxiliary signals' cost functions are chosen as $E_{i,\xi}(k) = \frac{1}{2}O_{i,\xi}^2(k)$ and $E_{i,\Gamma}(k) = \frac{1}{2}O_{i,\Gamma}^2(k)$. Then, the updating laws of $\hat{\xi}_i(k)$ and $\hat{\Gamma}_i(k)$ are derived as:

$$\begin{aligned} \hat{\xi}_i(k+1) &= \hat{\xi}_i(k) - \lambda_{i,\xi} [\hat{\xi}_i(k) + \hat{L}_i(k)] \\ \hat{\Gamma}_i(k+1) &= \hat{\Gamma}_i(k) - \lambda_{i,\Gamma} [\hat{\Gamma}_i(k) + \hat{L}_i(k)] \end{aligned} \quad (19)$$

where $\lambda_{i,\xi}$ and $\lambda_{i,\Gamma}$ are positive learning rates, define $\tilde{\xi}_i(k) = \hat{\xi}_i(k) - \xi_i$ and $\tilde{\Gamma}_i(k) = \hat{\Gamma}_i(k) - \Gamma_i$ as approximation errors.

Remark 1: Since $A_{il}, l = 1, \dots, n$ are known specified matrices, the virtual input $\alpha_{i,l}(k)$ for $\tilde{x}_{ril}(k)$ only contains the unknown item $\mathcal{F}_{i,l}(\tilde{x}_{i,n}(k))$, that is to say, $\alpha_{i,l}(k)$ is the function of $\tilde{x}_{i,n}(k)$.

C. Stability Analysis of the Virtual Learning Layer

The following theorem displays the performance analysis of the distributed reinforcement learning based cooperative learning strategy.

Theorem 1: Consider the uncertain tracking error system $\tilde{x}_{ri}(k)$ over a compact set. The virtual inputs are determined as $\hat{\alpha}_{i,1}(k)$, $\hat{\alpha}_{i,l}(k)$ and $\hat{v}_i(k)$ and the updating laws are specified as (5), (8), (12), (17) and (19), then all the signals of the closed-loop tracking error system $\tilde{x}_{ri}(k)$ and the cooperative learning errors are semiglobally uniformly ultimately bounded (SGUUB) once the design parameters satisfy $0 < \lambda_{i,j} < (1/\iota)$, $0 < \lambda_{i,c} < [1/(\iota\pi_i^2)]$, $0 < \lambda_{i,\xi} < 1$ and $0 < \lambda_{i,\Gamma} < 1$.

Proof: The procedure of the proof is given in the Appendix.

IV. OPTIMIZATION FOR DISTRIBUTED INPUT LAYER

This section realizes the learning task of the behaviors between the virtual objects and the leader with the data-based learning strategy. To reach consensus, we formulate a resulting deterministic optimization problem and resolve it effectively with the policy iteration algorithm.

A. Policy Iteration Algorithm

Previous section has realized the learning tasks between the original system (1) and the virtual objective (4). To complete the cooperative learning issue of the multi-agent system (1),

we shall realize the learning task of the behaviors between the virtual objects and the leader. According to Theorem 1, one has $\lim_{k \rightarrow \infty} \|x_{i,1}(k) - x_{ri,1}(k)\| = 0$. Let the approximation of the local neighbor tracking error of $e_i(k)$ be

$$e_{ri}(k) = \sum_{j \in \mathbb{N}_i} a_{ij} (x_{ir,1}(k) - x_{rj,1}(k)) + h_i (x_{ri,1}(k) - x_0(k)) \quad (20)$$

Similarly, the utility function is given as

$$\begin{aligned} \hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) &= e_{ri}^T(k) Q_{ii} e_{ri}(k) \\ &+ u_{ri}^T(k) R_{ii} u_{ri}(k) + \sum_{j \in \mathbb{N}_i} u_{r\mathbb{N}_i}^T(k) R_{ij} u_{r\mathbb{N}_i}(k). \end{aligned} \quad (21)$$

Use $\hat{r}_i(k)$ to represent $\hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k))$. And the corresponding long term performance index is

$$\begin{aligned} \hat{J}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) &= \sum_{l=k}^{\infty} \gamma^{l-k} \hat{r}_i(l) \\ &= \hat{r}_i(k) + \gamma \hat{J}_i(e_{ri}(k+1), u_{ri}(k+1), u_{r\mathbb{N}_i}(k+1)). \end{aligned}$$

Hence, one has a new system consisted of (4) and (20). If all the control policies are given admissible, then the value function of this new system can be defined as $\hat{\Theta}_i(e_{ri}(k)) = \sum_{l=k}^{\infty} \gamma^{l-k} \hat{r}_i(e_{ri}(l), u_{ri}(l), u_{r\mathbb{N}_i}(l)) = \hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) + \gamma \hat{\Theta}_i(e_{ri}(k+1))$. Then one has the discrete-time Bellman equation

$$\hat{\Theta}_i^*(e_{ri}(k)) = \hat{r}_i(e_{ri}(k), u_{ri}^*(k), u_{r\mathbb{N}_i}^*(k)) + \gamma \hat{\Theta}_i^*(e_{ri}(k+1)) \quad (22)$$

with $u_{ri}^*(k) = \operatorname{argmin}_{u_{ri}(k)} \{ \hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) + \gamma \hat{\Theta}_i^*(e_{ri}(k+1)) \}$, where $\hat{\Theta}_i^*(e_{ri}(k))$ and $u_{ri}^*(k)$ represent the optimal value function and the optimal input policy, respectively.

By employing the policy iteration algorithm, the value function and the input policy are updated iteratively, where the iteration index $p \rightarrow \infty$. The iteration procedure is given in Algorithm 1.

B. Convergence Analysis

In this section, according to Definition 1, the convergence analysis of policy iteration is given.

Lemma 2 [38]: For $p = 0, 1, \dots$, $\hat{\Theta}_i^p(e_{ri}(k))$ and $u_{ri}^p(k)$ are obtained by (23) and (24). If $u_{ri}^0(k)$ is admissible, then arbitrary policy obtained by (24) can stabilize the new system consisted of (4) and (20).

Theorem 2: For $p = 0, 1, \dots$, $\hat{\Theta}_i^p(e_{ri}(k))$ and $u_{ri}^p(k)$ are obtained by (23) and (24). If $u_{ri}^0(k)$ is admissible, then for $\forall e_{ri}(k) \in \mathbb{R}$, the value function $\hat{\Theta}_i^p(e_{ri}(k))$ is a monotonically non increasing sequence, i.e., $\hat{\Theta}_i^{p+1}(e_{ri}(k)) \leq \hat{\Theta}_i^p(e_{ri}(k))$.

Theorem 3: For $p = 0, 1, \dots$, $\hat{\Theta}_i^p(e_{ri}(k))$ and $u_{ri}^p(k)$ are obtained by (23) and (24). If $u_{ri}^0(k)$ is admissible, then we have the iterative value function $\hat{\Theta}_i^p(e_{ri}(k))$ and input law $u_{ri}^p(k)$ will converge to the optimal value function $\hat{\Theta}_i^*(e_{ri}(k))$ and the optimal policy $u_{ri}^*(k)$ as $p \rightarrow \infty$.

Algorithm 1: Policy Iteration Algorithm.**Initialization:**

Choose a randomly initial state $x_{ri}(0)$, a computation precision ε_r , an arbitrary admissible input $u_{ri}^0(k)$, and the max iteration of the computation $p_{\max}$;

Iteration:

- 1: Compute the corresponding value function $\hat{\Theta}_i^0(k)$ with the initial policy $u_{ri}^0(k)$:

$$\hat{\Theta}_i^0(k) = \hat{r}_i(e_{ri}(k), u_{ri}^0(k), u_{r\mathbb{N}_i}(k)) + \gamma \hat{\Theta}_i^0(e_{ri}(k+1));$$

- 2: Update the iterative policy by

$$u_{ri}^1(k) = \underset{u_{ri}(k)}{\operatorname{argmin}} \{ \hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) + \gamma \hat{\Theta}_i^0(e_{ri}(k+1)) \}.$$

- 3: Let $p = p + 1$, then construct the value function $\hat{\Theta}_i^p(e_{ri}(k))$ with

$$\hat{\Theta}_i^p(e_{ri}(k)) = \hat{r}_i(e_{ri}(k), u_{ri}^p(k), u_{r\mathbb{N}_i}^p(k)) + \gamma \hat{\Theta}_i^p(e_{ri}(k+1)). \quad (23)$$

- 4: Update the iterative policy by

$$u_{ri}^{p+1}(k) = \underset{u_{ri}(k)}{\operatorname{argmin}} \{ \hat{r}_i(e_{ri}(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) + \gamma \hat{\Theta}_i^p(e_{ri}(k+1)) \}. \quad (24)$$

- 5: If $\hat{\Theta}_i^{p-1}(e_{ri}(k)) - \hat{\Theta}_i^p(e_{ri}(k)) < \varepsilon_r$, go to step 7. Else 6;
- 6: If $p < p_{\max}$, go to step 3. Else go to step 8;
- 7: Output $\hat{\Theta}_i^p(e_{ri}(k))$ and $u_{ri}^p(k)$.
- 8: The optimization can't be achieved within $p_{\max}$ iterations.

The detailed proof processes of Theorem 2 and Theorem 3 can be found in [25], so they are omitted here to save space.

Give admissible control policies $u_{ri}(k)$ for $\forall i$ and define the value function of system (1) as

$$\begin{aligned} \Theta_i(e_i(k)) &= \sum_{l=k}^{\infty} \gamma^{l-k} r_i(e_i(l), u_{ri}(l), u_{r\mathbb{N}_i}(l)) \\ &= r_i(e_i(k), u_{ri}(k), u_{r\mathbb{N}_i}(k)) + \gamma \Theta_i(e_i(k+1)). \end{aligned} \quad (25)$$

Based on the results of Theorem 1, $\hat{\Theta}_i(e_{ri}(k))$ can be regarded as the estimation of $\Theta_i(e_i(k))$. According to (25), one has

$$\Theta_i^*(e_i(k)) = r_i(e_i(k), u_{ri}^*(k), u_{r\mathbb{N}_i}^*(k)) + \gamma \Theta_i^*(e_i(k+1)). \quad (26)$$

Theorem 4: Consider the multi-agent system (1), the weight updated laws (8), (12) and (17), as well as the auxiliary signals adaptive laws (19). Then, the tracking error $e_i(k)$ of system (1) converges to a small neighborhood around the origin and the

estimation of $\hat{\Theta}_i^*(e_{ri}(k))$ converges to $\Theta_i^*(e_i(k))$ with a small boundary layer.

Proof: Define a Lyapunov function:

$$V_{\Theta_i}(k) = \gamma^k \hat{\Theta}_i(e_{ri}(k)).$$

Hence, one has

$$\begin{aligned} \Delta V_{\Theta_i}(k) &= \gamma^{k+1} \hat{\Theta}_i(e_{ri}(k+1)) - \gamma^k \hat{\Theta}_i(e_{ri}(k)) \\ &= -\gamma^k \hat{r}_i(e_{ri}(k), u_{ri}^*(k), u_{r\mathbb{N}_i}^*(k)) \leq 0. \end{aligned} \quad (27)$$

Considering (27), one concludes that $\hat{\Theta}_i^*(e_{ri}(k))$ is stable, which yields $e_{ri}(k)$ converges to 0 when $k \rightarrow \infty$. We can further acquire the virtual system (4) and the leader will reach consensus with $\hat{\Theta}_i^*(e_{ri}(k))$.

According to Theorem 1, we have $x_i(k)$ well tracking the virtual objective $x_{ri}(k)$. With the result $e_{ri}(k) \rightarrow 0$, $e_i(k)$ will be convergent to 0 as well. Therefore the estimation of the optimal value function $\hat{\Theta}_i^*(e_{ri}(k))$ will be convergent to $\Theta_i^*(e_i(k))$ with a small boundary.

Remark 2: In this paper, we first make system (1) track the virtual objective with the designed adaptive weight laws. Thus, the actual input signal of the multi-agent system becomes $u_{ri}(k)$ with the output being $x_{i,1}(k)$. Then the solution of $u_{ri}(k)$ is solved by getting the results of the distributed input layer's Bellman equation.

Remark 3: To achieve the data-based cooperative learning optimization, we construct the above two-layer control framework. Unlike other traditional backstepping techniques for non-linear systems, the long term value function in the distributed layer contains not only the agent output data but also input signals of its own and neighbors'. This makes our proposed method more suitable for actual engineering systems to deal with energy consumption than the methods in [23], [39].

V. SIMULATION RESULTS

In this section, some numerical results are presented to illustrate the proposed cooperative learning method is effective.

A. Experimental Setting

To show the practical significance of our proposed method, we perform the simulation on a virtual multi-robot formation system which consists of one unmanned surface vessel (USV) leader and three USV followers. In such a multi-robot system, each USV can only communicate with its neighbors. The adjacency weights of the communication topology graph are given by $h_1 = 1$, $h_2 = 0$, $h_3 = 0$, $a_{12} = 0$, $a_{13} = 0$, $a_{21} = 1$, $a_{23} = 0$, $a_{31} = 1$, $a_{32} = 1$. To complete the formation task, all the USVs have to conduct cooperative learning to reach synchronization. The trajectories of the leader in x and y directions are given by $[x_0(k), y_0(k)]^T = [0.15 + 0.3\cos(k\pi/20.98), 0.15 - 0.3\sin(k\pi/20.98)]^T$. The dynamics of each USV are as follows: $x_{i,1}(k+1) = \frac{0.1x_{i,1}^2(k)}{1+x_{i,1}^2(k)} + b_{i,1}x_{i,2}(k)$, $x_{i,2}(k+1) = \frac{x_{i,2}^2(k)}{5+x_{i,1}^2(k)+x_{i,2}^2(k)} + 0.88u_{fi}(k) + d_{i,x}(k)$ with $y_{i,j}(k)$, $j = 1, 2$ sharing the same updating laws as $x_{i,j}(k)$. The external disturbances are

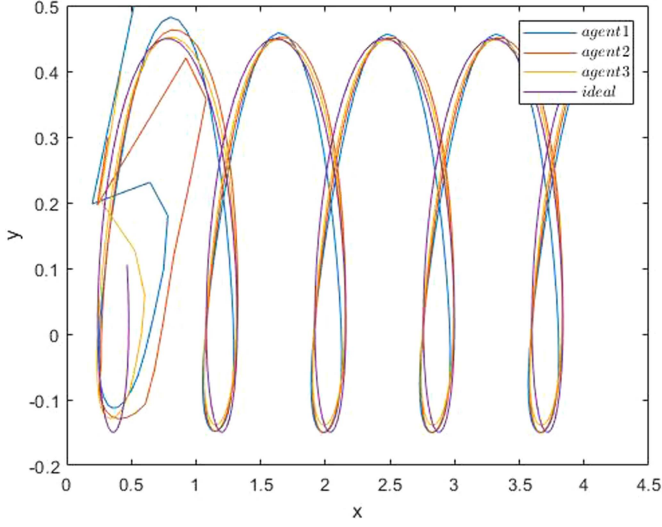
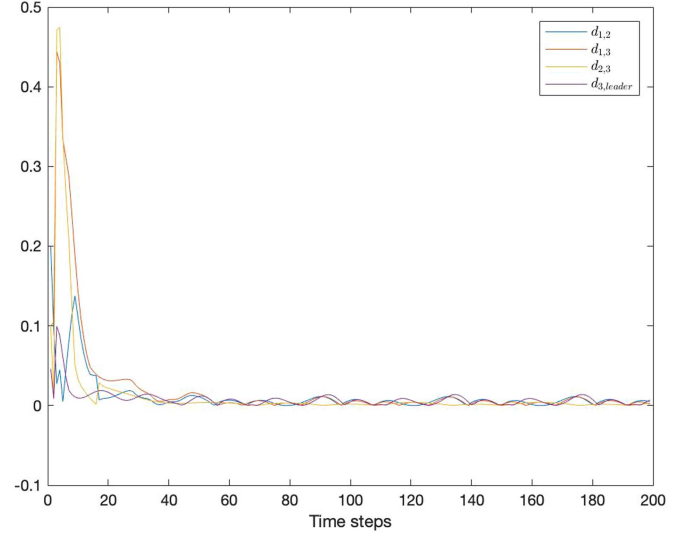


Fig. 2. Trajectories of three follower USVs and the leader.

Fig. 3. The distances $d_{i,j}$ between the USVs.

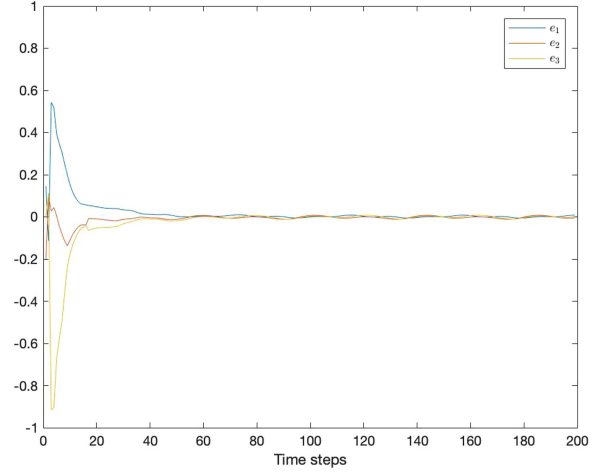
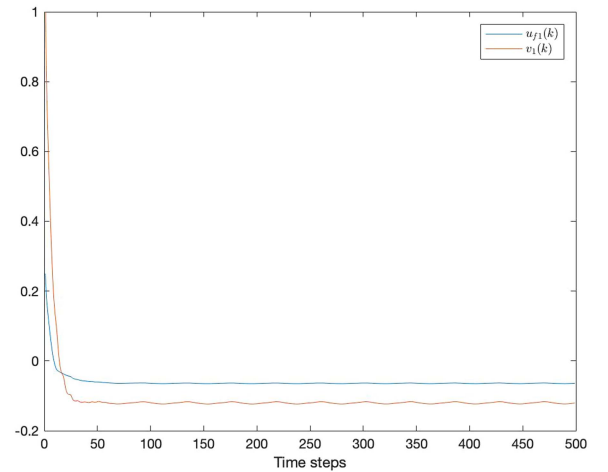
$d_{i,x}(k) = 0.152 \sin(k\pi/20.98 + 0.985) \cos(x_{i,1}(k)) + 0.13$ and $d_{i,y}(k) = 0.3 \sin(k\pi/20.98 + \pi/0.95) \cos(y_{i,1}(k)) + 0.2$. The parameters of the virtual model are chosen as $A_{11} = -0.13$, $A_{12} = [-0.36, 0.05]$, $b_{1,1} = 2.18$, $b_{1,2} = 0.88$, $A_{21} = -0.1$, $A_{22} = [-0.8, 1.1]$, $b_{2,1} = 0.9$, $b_{2,2} = 0.88$, $A_{31} = 0.1$, $A_{32} = [-0.35, 0.9]$, $b_{3,1} = 0.9$, and $b_{3,2} = 0.88$. For $i, j = 1, 2, 3$, Q_{ii} and R_{ij} are chosen as 1 and 0.5, respectively. $\gamma = 0.95$. The index threshold is chosen as $c_1 = 0.00002$. The initial states of the system are selected as $[x_1(0), y_1(0)] = [0.5, 0.1; 0.5, 0.1]^T$, $[x_2(0), y_2(0)] = [0.3, 0.2; 0.3, 0.2]^T$, $[x_3(0), y_3(0)] = [0.4, 0.2; 0.4, 0.2]^T$. $g_{i,r}$, $g_{i,l}$, $t_{i,r}$, $t_{i,l}$, $\eta_i(k)$ and $\delta_i(k)$ are randomly chosen in $[0, 1]$. $u_{r1}(0) = 0.3145$, $u_{r2}(0) = 0.2942$, $u_{r3}(0) = 0.3750$. $\xi_i(0)$ and $\Gamma_i(0)$ are both selected as 0.02. $\lambda_{i,c} = 0.000306$, $\lambda_{i,1} = 0.00506$, $\lambda_{i,2} = 0.0644$, $\lambda_{i,\xi} = 0.052$, $\lambda_{i,\Gamma} = 0.08652$, $\pi_i = 0.0009$. Note that the parameters which are not specially marked are the same in the x and y directions.

The desired formation trajectory is designed to make the USVs advance in the x direction in the form of curve. Thus, in addition to the control instructions given by the CPU, we make all the USVs keep the uniform linear motions in x direction at a speed of 0.001 for each time step.

For $i = 1, 2, 3$, choose the centers of Gaussian RBF NN for $\Phi_{i,1}(k)$ in $[-2 \times 2] \times [-2 \times 2]$, $[-2 \times 2] \times [-2 \times 2]$ for $L_i(k)$ and $\Phi_{i,2}(k)$ with widths $\rho = \sqrt{2}$. The initial weights of the critic and actor NNs are respectively chosen as $W_{i,c}(0) = [0.03, 0.03, \dots, 0.03]_{10 \times 1}^T$, $W_{i,1}(0) = [0.003, 0.003, \dots, 0.003]_{10 \times 1}^T$, and $W_{i,2}(0) = [0.3, 0.3, \dots, 0.3]_{10 \times 1}^T$. Note that $u_{ri}(0)$, $i = 1, 2, 3$ determines the initial control signal of the distributed control layer. According to the above discussions in Section IV, $u_{ri}(0)$ has to be admissible, otherwise, the multi-agent system will not be stable.

B. Results and Analysis

The simulation results are shown in Figs. 2–7. Fig. 2 shows the trajectories of three followers and the leader. From Fig. 2 it can

Fig. 4. Tracking errors $e_i(k)$ in the x direction.Fig. 5. Input $v_1(k)$ and fault-tolerant input $u_{f1}(k)$ in the x direction.

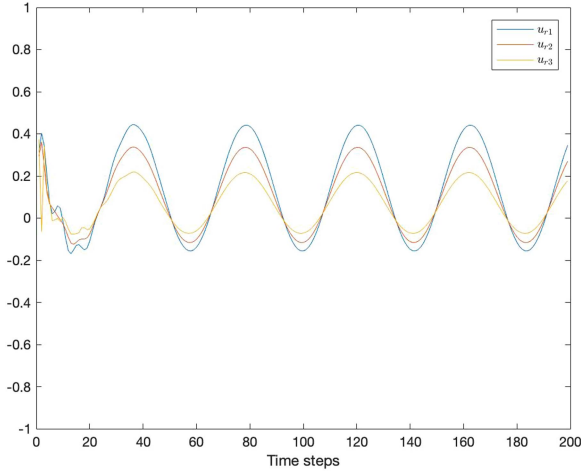


Fig. 6. Actual input $u_{T,i}(k)$ in the x direction.

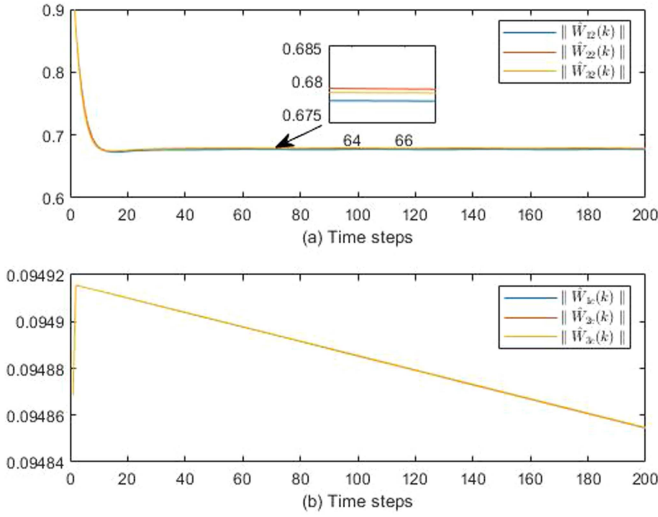


Fig. 7. Trajectories of $\|\hat{W}_{i,2}(k)\|$ and $\|\hat{W}_{i,c}(k)\|$ with $i = 1, 2, 3$ in x direction.

be seen the desired formation shape is established. The distances between all the neighbors are presented in Fig. 3. It is clear from Fig. 3 that the neighborhood distances fluctuate around 0. Fig. 4 describes the tracking errors between each agent and the leader. It shows that the tracking errors converge to small neighborhoods of zero using adaptive NNs. Fig. 5 shows the influence of the actuator fault and the dead zone on the virtual input. Fig. 6 shows the actual input of the system. The simulation results have proved that the cooperative learning method can compensate the influence of the system fault subjected to the actuator fault and the input dead zone.

During the learning process, trajectories of the norm of actor and critic NNs in the x direction are given in Fig. 7(a) and (b), respectively. Trajectories in the y direction are similar to the above results. $\hat{W}_{i,c}(k)$ for $\forall i$ represent the weights of the critic NNs which are used to assess the long term performance index, and $\hat{W}_{i,2}(k)$ for $\forall i$ are the weights of actor NNs which are used to make decisions in the virtual input layer. It can be seen from Fig. 7(a) and Fig. 5 the weight estimates $\hat{W}_{i,2}(k)$ begin to

converge around 35 time step and the virtual inputs $v_i(k)$ begin to stabilize. Then the system becomes stable and all the signals in the closed-loop system are SGUUB.

Figs. 3 and 4 show the trajectories of the tracking errors and sailing distances, which indicate that all the follower USVs can track the leader USV in a small gap. Due to the practical size of USVs, such a range of tracking distances is reasonable during voyaging. Thus, one can conclude that the multi-agent system has a good cooperative learning performance without prior knowledge of the model using the reinforcement learning strategy. Furthermore, compared to the existing methods using backstepping, the proposed algorithm applies the long term performance index with output and local input data, while others' only contain the error item. In addition, such a data-based method can reduce the communication cost and save resources to a certain degree. This simulation result also proves that the application in the field of multi-agent navigation is promising.

VI. CONCLUSION

A data-based reinforcement learning algorithm has been developed for the cooperative learning problem of nonlinear multi-agent systems. Specifically, the cooperative learning framework contains two layers with the introduction of the virtual learning objective. We designed the virtual adaptive input laws with the backstepping technique and critic-actor NNs, enabling an agent to perform its cooperative learning objective by selecting what action to perform. Meanwhile, all the state signals of the cooperative learning error systems are SGUUB. Actual input signals are obtained by using the policy iteration algorithm. In the future, our research topics will focus on the cooperative learning algorithms establishment for multi-agent systems in a more complex environment, such as stochastic noises, time-varying parameters, communication delays and so on.

APPENDIX A

THE PROOF OF THEOREM 1

Step 1: Define $\varsigma_{i,c}(k) = \tilde{W}_{i,c}^T(k)\Phi_{i,c}(k)$ and $\varsigma_{i,l}(k) = \tilde{W}_{i,l}^T(k)\Phi_{i,l}(k)$ for $l = 1, \dots, n$. Consider the Lyapunov function as follows:

$$V_{i,1}(k) = V_{i,11}(k) + V_{i,12}(k) + V_{i,13}(k) + V_{i,14}(k) \quad (28)$$

where

$$V_{i,11}(k) = \frac{\kappa_{i,11}}{3} e_{i,1}^2(k)$$

$$V_{i,12}(k) = \frac{\kappa_{i,w1}}{\lambda_{i,1}} \sum_{\epsilon=0}^{n-1} \tilde{W}_{i,1}^T(k_1 + \epsilon) \tilde{W}_{i,1}(k_1 + \epsilon)$$

$$V_{i,13}(k) = \frac{\kappa_{i,c}}{\lambda_{i,c}} \tilde{W}_{i,c}^T(k) \tilde{W}_{i,c}(k)$$

$$V_{i,14}(k) = 2\kappa_{i,c} \varsigma_{i,c}^2(k-1)$$

with $\kappa_{i,11}$, $\kappa_{i,c}$ and $\kappa_{i,w1}$ being positive constants.

Combining with the Cauchy-Schwarz inequality, one has

$$\Delta V_{i,11}(k) \leq \kappa_{i,11} b_{i,1}^2 e_{i,2}^2(k) + \kappa_{i,11} b_{i,1}^2 \varsigma_{i,1}^2(k_1)$$

$$\begin{aligned}
& + \kappa_{i,11} b_{i,1}^2 \bar{\varepsilon}_{i,1}^2 - \frac{\kappa_{i,11}}{3} e_{i,1}^2(k) \\
\Delta V_{i,12}(k) & \leq -\kappa_{i,w1} (1 - \iota \lambda_{i,1}) \left[\hat{W}_{i,1}^T(k_1) \Phi_{i,1}(k_1) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,w1} \varsigma_{i,1}^2(k_1) + 2\iota \kappa_{i,w1} [\bar{W}_{i,1} + \bar{W}_{i,c}]^2 + 2\kappa_{i,w1} \varsigma_{i,c}^2(k) \\
\Delta V_{i,13}(k) & \leq -\kappa_{i,c} (1 - \iota \lambda_{i,c} \pi_i^2) \left[\pi_i \hat{L}_i(k) - \hat{L}_i(k-1) + Q_i(k) \right]^2 \\
& - \kappa_{i,c} \pi_i^2 \varsigma_{i,c}^2(k) + 2\kappa_{i,c} \varsigma_{i,c}^2(k-1) + 2\iota \kappa_{i,c} [(1 + \pi_i) \bar{W}_{i,c} + 1]^2 \\
\Delta V_{i,14}(k) & = 2\kappa_{i,c} \varsigma_{i,c}^2(k) - 2\kappa_{i,c} \varsigma_{i,c}^2(k-1).
\end{aligned}$$

Since $\varepsilon_{i,c}$, $\varepsilon_{i,1}$, $W_{i,1}(k)$, $\Phi_{i,1}(k)$, $W_{i,c}(k)$ and $\Phi_{i,c}(k)$ are bounded, it follows that

$$\begin{aligned}
\Delta V_{i,1}(k) & \leq -\kappa_{i,w1} (1 - \iota \lambda_{i,1}) \left[\hat{W}_{i,1}^T(k_1) \Phi_{i,1}(k_1) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,c} (1 - \iota \lambda_{i,c} \pi_i^2) \times \left[\pi_i \hat{L}_i(k) - \hat{L}_i(k-1) + Q_i(k) \right]^2 \\
& - \frac{\kappa_{i,11}}{3b_{i,1}^2} e_{i,1}^2(k) - [\kappa_{i,w1} - \kappa_{i,11} b_{i,1}^2] \varsigma_{i,1}^2(k_1) \\
& - (\kappa_{i,c} \pi_i^2 - 2\kappa_{i,c} - 2\kappa_{i,w1}) \varsigma_{i,c}^2(k) \\
& + \kappa_{i,11} b_{i,1}^2 e_{i,2}^2(k) + M_{i,1} \tag{29}
\end{aligned}$$

where $M_{i,1} = \kappa_{i,11} b_{i,1}^2 \bar{\varepsilon}_{i,1}^2 + 2\iota \kappa_{i,w1} [\bar{W}_{i,1} + \bar{W}_{i,c}]^2 + 2\iota \kappa_{i,c} (\bar{W}_{i,c} (1 + \pi_i) + 1)^2$.

Step l ($2 \leq l \leq n-1$): Consider the Lyapunov function as follows:

$$V_{i,l}(k) = V_{i,l1}(k) + V_{i,l2}(k) \tag{30}$$

where $V_{i,l1}(k) = \frac{\kappa_{i,l1}}{3} e_{i,l}^2(k)$, $V_{i,l2}(k) = \frac{\kappa_{i,wl}}{\lambda_{i,l}} \sum_{\epsilon=0}^{n-l} \tilde{W}_{i,l}^T(k_l + \epsilon) \tilde{W}_{i,l}(k_l + \epsilon)$ with $\kappa_{i,l1}$ and $\kappa_{i,wl}$ being positive constants.

Then one can obtain

$$\begin{aligned}
\Delta V_{i,l1}(k) & \leq \kappa_{i,l1} b_{i,l}^2 e_{i,l+1}^2(k) + \kappa_{i,l1} b_{i,l}^2 \varsigma_{i,l}^2(k_l) \\
& + \kappa_{i,l1} b_{i,l}^2 \bar{\varepsilon}_{i,l}^2 - \frac{\kappa_{i,l1}}{3} e_{i,l}^2(k), \\
\Delta V_{i,l2}(k) & \leq -\kappa_{i,wl} (1 - \iota \lambda_{i,l}) \left[\hat{W}_{i,l}^T(k_l) \Phi_{i,l}(k_l) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,wl} \varsigma_{i,l}^2(k_l) + 2\kappa_{i,wl} \varsigma_{i,c}^2(k) + 2\iota \kappa_{i,wl} [\bar{W}_{i,l} + \bar{W}_{i,c}]^2.
\end{aligned}$$

It is obviously that

$$\begin{aligned}
\Delta V_{i,l}(k) & \leq -\kappa_{i,wl} (1 - \iota \lambda_{i,l}) \left[\hat{W}_{i,l}^T(k_l) \Phi_{i,l}(k_l) + \hat{L}_i(k) \right]^2 \\
& - (\kappa_{i,wl} - \kappa_{i,l1} b_{i,l}^2) \varsigma_{i,l}^2(k_l) - \frac{\kappa_{i,l1}}{3} e_{i,l}^2(k) \\
& + \kappa_{i,l1} b_{i,l}^2 e_{i,l+1}^2(k) + 2\kappa_{i,wl} \varsigma_{i,c}^2(k) + M_{i,l} \tag{31}
\end{aligned}$$

where $M_{i,l} = \kappa_{i,l1} b_{i,l}^2 \bar{\varepsilon}_{i,l}^2 + 2\iota \kappa_{i,wl} [\bar{W}_{i,l} + \bar{W}_{i,c}]^2$.

Step n : The candidate of Lyapunov function can be defined as

$$V_{i,n}(k) = V_{i,n1}(k) + V_{i,n2}(k) + V_{i,n3}(k) + V_{i,n4}(k) \tag{32}$$

where $V_{i,n1}(k) = \frac{\kappa_{i,n1}}{3} e_{i,n}^2(k)$, $V_{i,n2}(k) = \frac{\kappa_{i,wn}}{\lambda_{i,n}} \tilde{W}_{i,n}^T(k) \tilde{W}_{i,n}(k)$, $V_{i,n3}(k) = \frac{\kappa_{i,\xi}}{\lambda_{i,\xi}} \tilde{\xi}_i^2(k)$, $V_{i,n4}(k) = \frac{\kappa_{i,\Gamma}}{\lambda_{i,\Gamma}} \tilde{\Gamma}_i^2(k)$ with $\kappa_{i,n1}$, $\kappa_{i,wn}$, $\kappa_{i,\xi}$ and $\kappa_{i,\Gamma}$ being positive constants.

According to (17) and use the Cauchy-Schwarz inequality, one has

$$\begin{aligned}
\Delta V_{i,n1} & \leq \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 (\varsigma_{i,n}^2(k_n) + \bar{\varepsilon}_{i,n}^2 + \tilde{\xi}_i^2(k) \\
& + \tilde{\Gamma}_i^2(k) + 2\xi_i^2 + 2\Gamma_i^2) + \frac{2\kappa_{i,n1}}{3} \bar{d}^2 - \frac{\kappa_{i,n1}}{3} e_{i,n}^2(k) \\
\Delta V_{i,n2} & \leq -\kappa_{i,wn} (1 - \iota \lambda_{i,n}) \left[\hat{W}_{i,n}^T(k_n) \Phi_{i,n}(k_n) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,wn} \varsigma_{i,n}^2(k_n) + 2\iota \kappa_{i,wn} [\bar{W}_{i,n} + \bar{W}_{i,c}]^2 \\
& + 2\kappa_{i,wn} \varsigma_{i,c}^2(k) \\
\Delta V_{i,n3} & \leq -\kappa_{i,\xi} (1 - \lambda_{i,\xi}) \left[\hat{\xi}_i(k) + \hat{L}_i(k) \right]^2 - \kappa_{i,\xi} \tilde{\xi}_i^2(k) \\
& + 2\kappa_{i,\xi} [\xi_i + \bar{W}_{i,c}]^2 + 2\kappa_{i,\xi} \varsigma_{i,c}^2(k) \\
\Delta V_{i,n4} & \leq -\kappa_{i,\Gamma} (1 - \lambda_{i,\Gamma}) \left[\hat{\Gamma}_i(k) + \hat{L}_i(k) \right]^2 - \kappa_{i,\Gamma} \tilde{\Gamma}_i^2(k) \\
& + 2\kappa_{i,\Gamma} [\Gamma_i + \bar{W}_{i,c}]^2 + 2\kappa_{i,\Gamma} \varsigma_{i,c}^2(k).
\end{aligned}$$

It follows that

$$\begin{aligned}
\Delta V_{i,n}(k) & \leq -\kappa_{i,wn} (1 - \iota \lambda_{i,n}) \\
& \left[\hat{W}_{i,n}^T(k_n) \Phi_{i,n}(k_n) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,\xi} (1 - \lambda_{i,\xi}) \left[\hat{\xi}_i(k) + \hat{L}_i(k) \right]^2 \\
& - \kappa_{i,\Gamma} (1 - \lambda_{i,\Gamma}) \left[\hat{\Gamma}_i(k) + \hat{L}_i(k) \right]^2 \\
& - \left(\kappa_{i,wn} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \varsigma_{i,n}^2(k_n) \\
& - \left(\kappa_{i,\xi} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \tilde{\xi}_i^2(k) \\
& - \left(\kappa_{i,\Gamma} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \tilde{\Gamma}_i^2(k) \\
& + (2\kappa_{i,wn} + 2\kappa_{i,\xi} + 2\kappa_{i,\Gamma}) \varsigma_{i,c}^2(k) \\
& - \frac{\kappa_{i,n1}}{3} e_{i,n}^2(k) + M_{i,n} \tag{33}
\end{aligned}$$

where $M_{i,n} = \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 (\bar{\varepsilon}_{i,n}^2 + 2\xi_i^2 + 2\Gamma_i^2) + \frac{2\kappa_{i,n1}}{3} \bar{d}^2 + 2\iota \kappa_{i,wn} [\bar{W}_{i,n} + \bar{W}_{i,c}]^2 + 2\kappa_{i,\xi} [\xi_i + \bar{W}_{i,c}]^2 + 2\kappa_{i,\Gamma} [\Gamma_i + \bar{W}_{i,c}]^2$.

The global Lyapunov function can be defined as

$$V(k) = \sum_{i=1}^N \sum_{j=1}^n V_i(k). \tag{34}$$

From the above analysis one can obtain

$$\Delta V(k)$$

$$\begin{aligned}
&\leq -\sum_{i=1}^N \sum_{j=1}^n \kappa_{i,wj} (1 - \lambda_{i,j}) \left[\hat{W}_{i,j}^T(k_j) \Phi_{i,j}(k_j) + \hat{L}_i(k) \right]^2 \\
&\quad - \sum_{i=1}^N \kappa_{i,c} (1 - \lambda_{i,c} \pi_i^2) \left[\pi_i \hat{L}_i(k) - \hat{L}_i(k-1) + Q_i(k) \right]^2 \\
&\quad - \sum_{i=1}^N (\kappa_{i,c} \pi_i^2 - 2\kappa_{i,c} - 2\kappa_{i,\xi} - 2\kappa_{i,\Gamma}) \\
&\quad - 2 \sum_{j=1}^n \kappa_{i,wj} \varsigma_{i,c}^2(k) - \sum_{i=1}^N \frac{\kappa_{i,11}}{3} e_{i,1}^2(k) \\
&\quad - \sum_{i=1}^N \sum_{j=2}^n \left(\frac{\kappa_{i,j1}}{3} - \kappa_{i,(j-1)1} b_{i,j-1}^2 \right) e_{i,j}^2(k) \\
&\quad - \sum_{i=1}^N \sum_{j=1}^{n-1} (\kappa_{i,wj} - \kappa_{i,j1} b_{i,j}^2) \varsigma_{i,j}^2(k_j) \\
&\quad - \sum_{i=1}^N \left(\kappa_{i,wn} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \varsigma_{i,n}^2(k_n) \\
&\quad - \sum_{i=1}^N \kappa_{i,\xi} (1 - \lambda_{i,\xi}) \left[\hat{\xi}_i(k) + \hat{L}_i(k) \right]^2 \\
&\quad - \sum_{i=1}^N \kappa_{i,\Gamma} (1 - \lambda_{i,\Gamma}) \left[\hat{\Gamma}_i(k) + \hat{L}_i(k) \right]^2 \\
&\quad - \sum_{i=1}^N \left(\kappa_{i,\xi} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \tilde{\xi}_i^2(k) \\
&\quad - \sum_{i=1}^N \left(\kappa_{i,\Gamma} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \tilde{\Gamma}_i^2(k) + M \quad (36)
\end{aligned}$$

Clearly, the parameters can be selected as $\kappa_{i,j1} > 3\kappa_{i,(j-1)1} b_{i,j}^2$, with $j = 2, \dots, n$, $\kappa_{i,wj} > \kappa_{i,j1} b_{i,j}^2$ with $j = 1, \dots, n-1$, $\kappa_{i,wn} > \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2$, $\kappa_{i,\xi} > \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2$, $\kappa_{i,\Gamma} > \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2$ and $\kappa_{i,c} \pi_i^2 > 2\kappa_{i,c} + 2\kappa_{i,\xi} + 2\kappa_{i,\Gamma} + 2 \sum_{j=1}^n \kappa_{i,wj}$.

It is concluded that $\Delta V(k) < 0$ as long as $|e_{i,1}(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,11}/3}}$ or $|e_{i,j}(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,j1}/3 - \kappa_{i,(j-1)1} b_{i,j-1}^2}}$, $j = 2, \dots, n$, or $|\varsigma_{i,j}(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,wj} - \kappa_{i,j1} b_{i,j}^2}}$, $j = 1, \dots, n-1$ or $|\varsigma_{i,n}(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,wn} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2}}$ or $|\tilde{\xi}_i(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,\xi} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2}}$ or $|\tilde{\Gamma}_i(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,\Gamma} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2}}$ or $|e_{i,c}(k)| > \frac{\sqrt{M}}{\sqrt{\kappa_{i,c} \pi_i^2 - 2\kappa_{i,c} - 2\kappa_{i,\xi} - 2\kappa_{i,\Gamma} - 2 \sum_{j=1}^n \kappa_{i,wj}}}$. As a result, one can conclude that all signals are SGUUB.

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where $M = \sum_{i=1}^N M_i$.

By selecting the proper parameters $\lambda_{i,j}$, $\lambda_{i,c}$, $\lambda_{i,\xi}$ and $\lambda_{i,\Gamma}$, one has

$$\begin{aligned}
\Delta V(k) &\leq -\sum_{i=1}^N (\kappa_{i,c} \pi_i^2 - 2\kappa_{i,c} - 2\kappa_{i,\xi} - 2\kappa_{i,\Gamma}) \\
&\quad - 2 \sum_{j=1}^n \kappa_{i,wj} \varsigma_{i,c}^2(k) - \sum_{i=1}^N \frac{\kappa_{i,11}}{3} e_{i,1}^2(k) \\
&\quad - \sum_{i=1}^N \sum_{j=2}^n \left(\frac{\kappa_{i,j1}}{3} - \kappa_{i,(j-1)1} b_{i,j-1}^2 \right) e_{i,j}^2(k) \\
&\quad - \sum_{i=1}^N \sum_{j=1}^{n-1} (\kappa_{i,wj} - \kappa_{i,j1} b_{i,j}^2) \varsigma_{i,j}^2(k_j) \\
&\quad - \sum_{i=1}^N \left(\kappa_{i,wn} - \frac{16\kappa_{i,n1}}{3} b_{i,n}^2 \bar{\eta}^2 \bar{g}^2 \right) \varsigma_{i,n}^2(k_n)
\end{aligned}$$

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